Tejas Vikas Mundada

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.46
Takhatmal English High School	Class XII (Maharashtra State Board)	2024	95.17%
Takhatmal English High School	Class X (Maharashtra State Board)	2022	96.40%

SCHOLASTIC ACHIEVEMENTS

- Secured **Department Rank 3** in Mathematics and Computing for two consecutive academic years (based on CGPA)
- Ranked as **Specialist** on codeforces with **max rating of 1570** and **solved 500+** competitive programming problems (Profile)
- Selected for **Indian National Chemistry Olympiad (INChO-2024)** by ranking among the **top 531** candidates (out of 44363) nationwide in National Standard Examination in Chemistry (**NSEC-2023**)
- Achieved **state-topper** recognition in National Standard Examination in Physics (**NSEP-2023**) for having been placed among **Top 1%** of the candidates

PROJECTS

Custom SQL Database Engine | C++

- Built a persistent **SQL database engine in C++** featuring a custom **CLI REPL** to parse and execute core **CRUD** operations
- **Implemented a B+ tree index** mapping primary keys to physical disk offsets, optimizing lookups to O(logN) time complexity
- Engineered an **LRU Cache Buffer Pool** using a hash map and doubly linked list, drastically **minimizing disk I/O operations**
- Designed **binary serialization** via std::fstream to write structured data directly to disk for strict persistence

Interactive Graham Scan Visualizer | C++, Raylib **|** DSA Lab Course Project (Prof. J. Balasubramaniam)

- Developed an **interactive tool** to compute and animate the **Convex Hull of 2D points** using **Graham Scan** algorithm
- Utilized a **stack-based approach** to dynamically track the hull vertices, achieving an optimal O(NlogN) time complexity
- Rendered **real-time animations using Raylib** to visually demonstrate point selection, stack operations and hull construction

Connect Four AI Engine | C++

- **Developed a game engine** featuring robust game-state management, move generation, and win-detection logic **using OOP**
- **Implemented game-tree search algorithm** with custom heuristic evaluation and forced-move detection to find optimal moves
- **Optimized execution time** using backtracking and performance profiling (chrono), ensuring efficient state-space exploration

Momentum-Tilted Risk Parity Backtester | Python, NumPy, Pandas, yfinance

- Engineered a **robust portfolio backtester** incorporating strict look-ahead bias prevention and **0.1% transaction cost** model
- Implemented **Risk Parity framework** combining 30-day inverse volatility strategy with a 90-day momentum penalty

tilt
- Minimized portfolio churn and execution friction by enforcing a **5% tolerance-band rebalancing threshold**
- Achieved **1.54 Sharpe ratio**, **50.4% return** and minimized **max drawdown to -5.6%** over a 2-year backtest on 10 BSE assets

Statistical and Data Analysis | Python **|** Course Project (Prof. Anurabha Majumdar)

- Processed real-world datasets using Python to build frequency tables, histograms, and box plots to extract statistical insights
- Modelled continuous, exponential, and discrete datasets to compute point estimations using MLE and validate CLT
- Executed one-sample t-tests, chi-square contingency tests, and linear regression models to evaluate variable dependencies

SKILLS

- **Programming Languages:** C, C++, Python
- **Tools and Libraries:** NumPy, Pandas, Matplotlib, PIL (Pillow), Raylib, LaTeX

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Data Structures and Applications, Data Structures and Applications Lab, Algorithms, Theory of Computation, Database Management Systems-1, Compilers-1
- **Mathematics:** Probability Theory, Applied Statistics, Introduction to Number Theory, Linear Algebra, Ordinary Differential Equations, Convex Optimization, Transform Techniques

POSITIONS OF RESPONSIBILITY

Web Coordinator | Inter IIT Sports Meet 2025

- Developed **Python automation scripts** using **PIL (Pillow)** to parse CSV data and overlay text onto template images
- **Automated the asset generation** for match schedules, results, ID cards, and certificates, **for 1900+ participants**
- Optimized operational workflow by **eliminating manual data entry**, significantly reducing project turnaround time

Finance Coordinator | Finance and Consulting Club, IIT Hyderabad

- Conducted workshops and sessions on **stock markets**, **technical analysis**, and **fundamental analysis** for students
- **Organized Investomania**, a **finance competition** in the annual flagship event - **Horizon**, managing end-to-end execution, problem statement design, and logistics in a 3-member team

Teaching Assistant | Series of Functions | Prof. J. Balasubramaniam

Class Representative, Mathematics and Computing | **Academic Year 2026-27**

EXTRACURRICULAR

- National Sports Organisation (NSO) Cricket Player
- **Winner** of the Institute Cricket League (**ICL-12**) and awarded **Man of the Match** in the **Final**
- Secured **1st position** in the **Campus Case Challenge** organized by Finance and Consulting Club, IIT Hyderabad
- Secured **Rank 750** in the **Prelims Round** of **IICPC CodeFest 2026** among 13000+ participants